

JAVA PROGRAMMIN G PROJECT PRESENTATION



KG COLLEGE OF ARTS AND SCIENCE
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Project Title: Bank Account Management System with Deposit, Withdrawal and Transaction History

Course: Java Programming / Mini Project

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JAVA PROGRAMMING PROJECT PRESENTATION

INTRODUCTION

This project is a Java-based application to manage bank accounts efficiently.

It allows users to deposit, withdraw money, and view transaction history.

Helps overcome limitations of manual banking records.

Useful for tracking account balance and transactions in one place.

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OBJECTIVE

Main goal of the project :

- To develop a bank account management system using Java

- To implement deposit and withdrawal functionalities

- To maintain transaction history

- To apply object-oriented programming concepts

- To improve programming and problem-solving skills

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PROBLEM STATEMENT

Managing bank transactions manually is difficult and error-prone

Difficult to track deposits, withdrawals, and balance accurately

No proper record of transaction history

This project provides a digital solution for easy bank management ↓

SYSTEM REQUIREMENTS

System Requirements

Hardware Requirements:

Processor: Intel i3 or higher

RAM: 4 GB minimum

Storage: 500 MB free space

Software Requirements:

OS: Windows / Linux

Programming Language: Java

IDE: Eclipse / IntelliJ / NetBeans

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SYSTEM DESIGN

User Interface
Account Management
Transaction Module
History Management
Include:
Flowchart
Block Diagram
Algorithm

MODULE DESCRIPTION

InputModule:

Accepts user details like account number, deposit/withdraw amount

Processing Module:

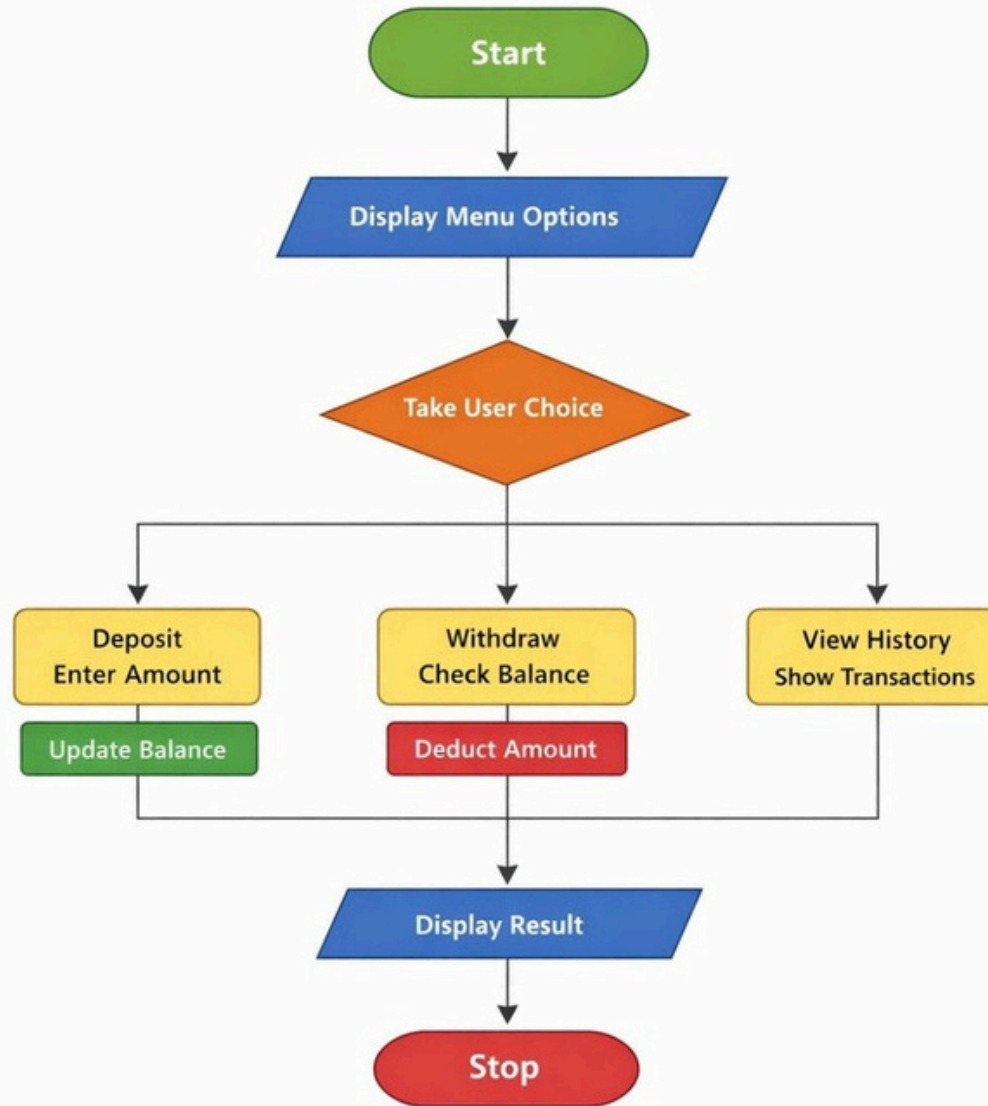
Handles deposit, withdrawal, and balance calculation

Output Module:

Displays account balance and transaction history

Transaction History Module:

Stores all transactions using ArrayList or file



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IMPLEMENTATION

Techniques used:
Classes and Objects (oops
concepts)
Array List for storing transactions
Loops and conditional statements
Basic file handling (optional)

```
import java.util.*;

class BankAccount {
    double balance;
    ArrayList<String> history = new ArrayList<>();

    BankAccount(double initialBalance) {
        balance = initialBalance;
    }

    void deposit(double amount) {
        balance += amount;
        history.add("Deposited: " + amount);
        System.out.println("Amount Deposited Successfully!");
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance -= amount;
            history.add("Withdrawn: " + amount);
            System.out.println("Amount Withdrawn Successfully!");
        } else {
            System.out.println("Insufficient Balance!");
        }
    }
}
```

```
        void showBalance() {
            System.out.println("Current Balance: " + balance);
        }

        void showHistory() {
            System.out.println("Transaction History:");
            for (String h : history) {
                System.out.println(h);
            }
        }
    }

    public class BankManagement {
        public static void main(String[] args) {
            Scanner sc = new Scanner(System.in);
            BankAccount acc = new BankAccount(1000);

            while (true) {
                System.out.println("\n1. Deposit\n2. Withdraw\n3. Balance\n4. History\n5. Exit");
                int choice = sc.nextInt();
                switch (choice) {
                    case 1:
                        System.out.print("Enter amount: ");
                        acc.deposit(sc.nextDouble());
                        break;
                    case 2:
                        System.out.print("Enter amount: ");
                        acc.withdraw(sc.nextDouble());
                        break;
```



```
case 3:
acc.showBalance();
    break;
case 4:
acc.showHistory();
    break;
case 5:
System.exit(0);
    }
    }
    }
    }
```

Menu display screen

Deposit confirmation screen

Withdrawal confirmation screen

Transaction history screen

Bank Account Management System

- 1. Deposit
- 2. Withdraw
- 3. Balance
- 4. History
- 5. Exit

Enter your choice: 1

Main Menu Display

Bank Account Management System

1. Deposit
2. Withdraw
3. Balance
4. History
5. Exit

Enter your choice: 1

Enter amount: 500

Amount Deposited Successfully!

Bank Account Management System

1. Deposit
2. Withdraw
3. Balance
4. History

Enter your choice: 2

Enter amount 300

Amount Withdrawn Successfully!

Withdraw Amount

Bank Account Management System

1. Deposit
2. Withdraw
3. Balance
4. History

Enter your choice: 4

Transsction History:
Deposited: 500

Transaction History

RESULTS & DISCUSSION

Successfully managed bank transactions

Accurate deposit and withdrawal operations

Maintained transaction history effectively

Challenges faced:

Handling user input

Managing transaction records

Solutions:

Used ArrayList for storing history

Applied proper logic and conditions

CONCLUSION & FUTURE ENHANCEMENT

Conclusion:

The project meets all objectives

Demonstrates Java programming and OOP concepts

Provides an efficient bank management system

Future Enhancements:

Add GUI using Java Swing or JavaFX

Use database (MySQL) for storage

Add login/security features

Online banking features

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REFERENCES / ACKNOWLEDGEMENT

References:

Java Programming - E. Balagurusamy

Oracle JavaDocumentation

Online tutorials

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THANK YOU